Body weight status and cardiovascular risk factors in adults by frequency of candy consumption

Background Limited information is available regarding the impact of candy consumption on health. The purpose of this study was to investigate associations between typical frequency of candy consumption and body weight status and select cardiovascular risk factors among adults in the United States. Methods Using data collected in the 2003–2006 National Health and Nutrition Examination Surveys (NHANES), adults were categorized as infrequent ($\leq$ 3 eating occasions [EO]/month), moderate ($>$ 3 EO/month and $\leq$ 3.5 EO/week), or frequent ($>$ 3.5 EO/week) candy consumers based on the combined frequency of chocolate and other candy consumption over the previous 12 months. Weight and adiposity status were analyzed using logistic regression models, and blood pressure, lipids, and insulin sensitivity were analyzed using linear regression models. Models were adjusted for age, sex and race/ethnicity, and also for additional covariates with potential associations with the outcomes. Appropriate statistical weights were used to yield results generalizable to the US population. Results Frequency of candy consumption was not associated with the risk of obesity, overweight/obesity, elevated waist circumference, elevated skinfold thickness, blood pressure, low density lipoprotein (LDL) or high density lipoprotein (HDL) cholesterol, triglycerides, or insulin resistance. Increased frequency of candy consumption was associated with higher energy intakes and higher energy adjusted intakes of carbohydrates, total sugars and added sugars, total fat, saturated fatty acids and monounsaturated fatty acids ($p < 0.05$), and lower adjusted intakes of protein and cholesterol ($p < 0.001$). Conclusions Increased frequency of candy consumption among adults in the United States was not associated with objective measures of adiposity or select cardiovascular risk factors, despite associated dietary differences. Given the cross-sectional study design, however, it cannot be concluded that candy consumption does not cause obesity or untoward levels of cardiovascular risk markers. The lack of an association between frequency of candy consumption and cardiovascular risk factors could be due to reduced intake of candy among the overweight due to dieting or a health professional's recommendations. Additionally, it is important to note that the analysis was based on frequency of candy consumption and not amount of candy consumed. Longitudinal studies are needed to confirm the lack of associations between frequency of candy consumption and cardiovascular risk factors.